Ezra Kahn

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Education

Carnegie Mellon University Pittsburgh, PA
B.S. in Mechanical Engineering; Minor in Robotics; Technical GPA: 3.5/4.0 *May 2028*
Selected Coursework: Engineering Statistics and Quality Control; Electronics for Sensing and Actuation; Dynamics; Mechanics; Fluid Mechanics; Thermodynamics; Fundamentals of Computer Science; Manual Machining; Human-Robot Interaction

Experience

MetaMobility Lab – Undergraduate Mechanical Researcher January 2026 – Present

- Designed custom adjustable pelvis, thigh, and shin IMU mounts and compact electronics packaging for a CAN-based distributed computing architecture, maintaining comparable sensor position and repeatable initial orientation across users with different body geometries while enabling robust communication between knee and hip exoskeleton controllers
- Designed and fabricated a custom thigh bar with a prismatic-rail interface for a hip exoskeleton, reducing assembly weight, improving torque transfer, and adding an additional degree of freedom
- Developed a custom strap attachment to minimize motor motion during gait, balancing stiffness, comfort, adjustability, and manufacturability for wearable robotics hardware

Dogwood Brands PE – Software Engineering Associate May 2025 – January 2026

- Automated daily KPI reporting across 60+ Supercuts locations by building Python and advanced Excel workflows for operational data analysis
- Delivered real-time product comparisons by developing a Python web scraper and contributing to backend data-processing and reporting systems
- Streamlined recruitment workflows and improved candidate engagement by developing an AI-powered HR chatbot

New York Stem Cell Foundation – Engineering Intern June 2023 – August 2023

- Designed an automated gas-pressure control prototype to replace manual regulators for critical stem-cell storage infrastructure
- Built a functional prototype by modeling components in SolidWorks, analyzing hardware datasheets, and implementing control logic
- Supported system integration and data visibility by digitizing monitoring of pressure-control behavior and infrastructure status

Projects

Carnegie Mellon Racing – Structures, Electrical Integration October 2024 – Present

- Model battery-enclosure components in SolidWorks for an electric Formula SAE racecar, satisfying structural packaging and electrical integration requirements
- Collaborate across Structures and Electrical Integration to align enclosure geometry, component packaging, and system integration requirements

3-DOF Robotic Arm – Personal Project June 2026 – Present

- Designed a 3-DOF robotic arm and gripper in SolidWorks, applying mass-property analysis and torque calculations to size servos for lightweight manipulation
- Verified joint motion and CAD alignment by building a ROS 2 Jazzy model with URDF/Xacro, TF, custom STL meshes, and RViz

Avenues Robotics Team – Mechanical Subteam Lead 2021 – 2024

- Led mechanical design, fabrication, and iteration for competition robots that qualified for world championships

Technical Skills

Programming & Data Analysis : Python, MATLAB, C++; Excel (advanced)
CAD & Mechanical Design : SolidWorks, Autodesk Fusion 360
Robotics & Hardware : ROS 2, URDF/Xacro, RViz; Arduino, IMUs, sensors, actuators
Manufacturing & Fabrication : 3D printing, waterjet cutting, manual machining, mill, lathe

Activities

Cello – Mannes Preparatory School Orchestra and Avenues High School Orchestra 2015 – 2024
Chelsea Piers Gymnastics – Nationally Ranked Level 10 Gymnast and Team Captain 2010 – 2024
myFace – Spokesperson and Fundraiser for People with Craniofacial Differences February 2020 – Present